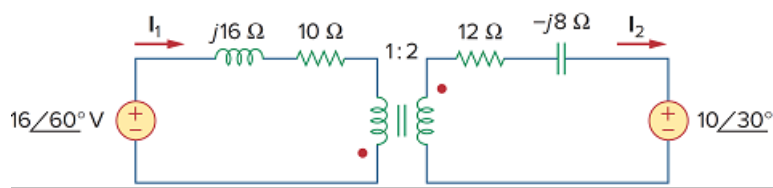


Problem

(a) Find I_1 and I_2 in the circuit of Fig. 13.111 below.

(b) Switch the dot on one of the windings. Find I_1 and I_2 again.

Figure 13.111

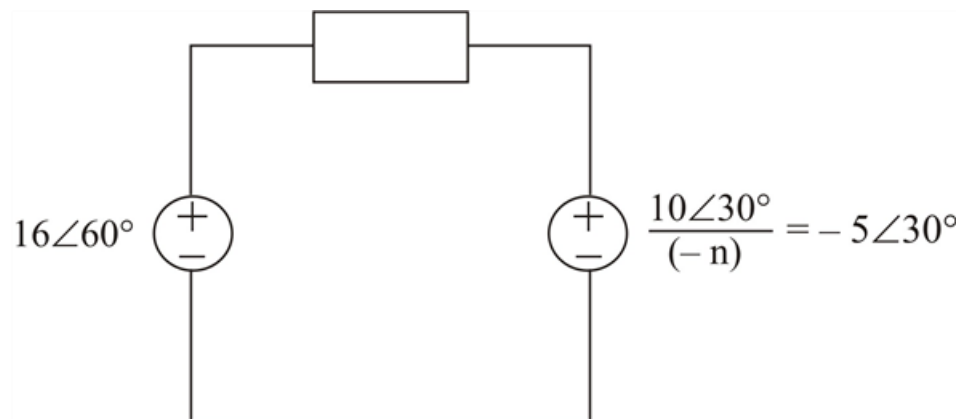


Step-by-step solution

Step 1 of 4

(A) Reflecting the secondary circuit to primary, we have the circuit shown below.

Step 2 of 4



Step 3 of 4

$$Z_{in} = 10 + j16 + \left(\frac{1}{4}\right)(12 - j8) = 13 + j14$$

$$-16\angle 60^\circ + Z_{in} I_1 - 5\angle 30^\circ = 0$$

Or,

$$I_1 = \frac{(16\angle 60^\circ + 5\angle 30^\circ)}{(13 + j14)}$$

Hence,

$$I_1 = 1.072\angle 5.88^\circ \text{ A}$$

$$I_2 = -0.5I_1 = 0.536\angle 185.88^\circ \text{ A}$$

Step 4 of 4

(B) Switching a dot will not effect Z_{in} but will effect I_1 & I_2 .

$$I_1 = \frac{(16\angle 60^\circ - 5\angle 30^\circ)}{(13 + j14)} = 0.625\angle 25^\circ A \quad \&$$

$$I_2 = 0.5I_1 = 0.3125\angle 25^\circ A \quad .$$